

PILATHARA CO-COPERATIVE ARTS & SCIENCE COLLEGE



**PILATHARA KANNUR-670504
(AFFILIATED TO KANNUR UNIVERSITY)**

PRACTICAL RECORD

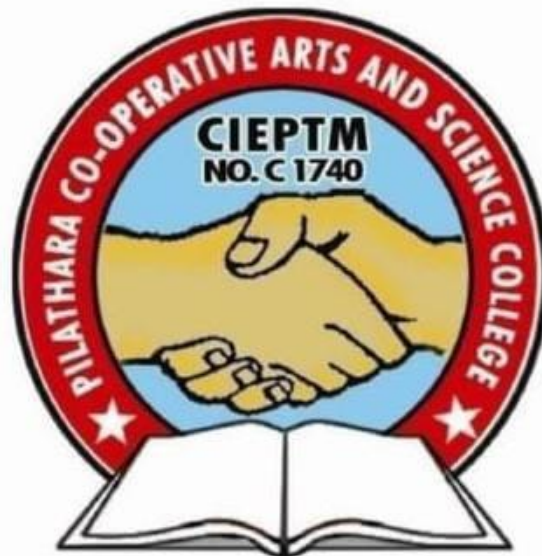
PYTHON

NAME :

REG.NO :

SEMESTER :

PILATHARA CO-COPERATIVE ARTS & SCIENCE COLLEGE



PILATHARA KANNUR- 670504

(AFFILIATED TO KANNUR UNIVERSITY)

PRACTICAL RECORD

CERTIFICATE

**CERTIFIED THAT THIS IS A BONAFIDE RECORD OF THE ORIGINAL WORK
DONE BY REG.NO.....
OF SECOND BCA IN THE DATA STRUCTURE & DBMS DURING THE YEAR
2022-2025.**

EXAMINERS:

LECTURER IN CHARGE:

1.

2.

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PROGRAM 1

AIM:

a program Write to find the largest from the list of numbers

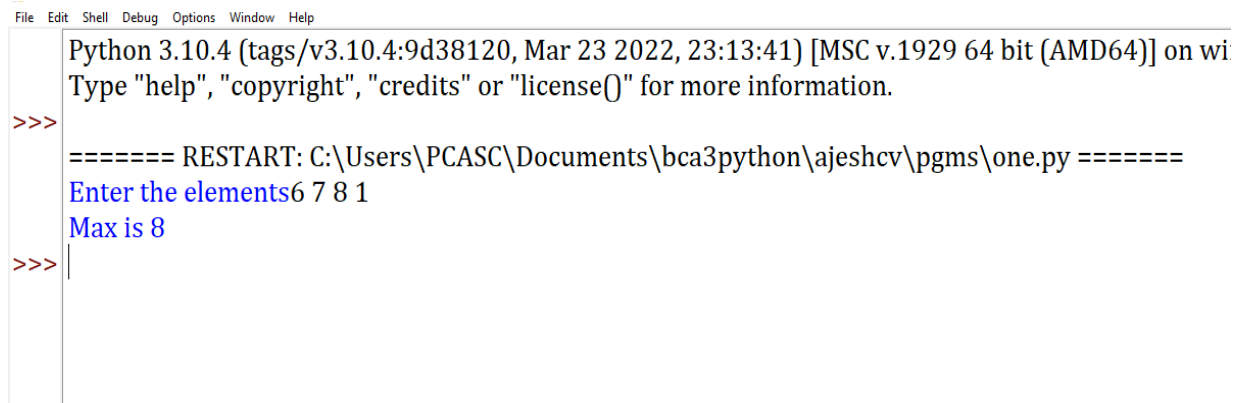
PROGRAM:

```
a=input("Enter the elements")
```

```
li=list(map(int,a.split()))
```

```
print("Max is",max(li))
```

OUTPUT:



```
File Edit Shell Debug Options Window Help
Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (AMD64)] on wi
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:\Users\PCASC\Documents\bca3python\ajeshcv\pgms\one.py =====
Enter the elements6 7 8 1
Max is 8
>>> |
```

PROGRAM 2

AIM:

Write a program to find the prime number in a list of numbers

PROGRAM:

```
str=input("Enter the elements")
```

```
m=list(map(int,str.split()))
```

```
p=[]
```

```
for i in m:
```

```
    c=0
```

```
    for j in range(1,i):
```

```
        if i%j==0:
```

```
            c+=1
```

```
    if c==1:
```

```
        p.append(i)
```

```
print("Prime numbers are",p)
```

OUTPUT:

```
File Edit Shell Debug Options Window Help
Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (AMD64)] on w
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:\Users\PCASC\Documents\bca3python\ajeshcv\pgms\two.py =====
Enter the elements4 6 1 7
Prime numbers are [7]
>>> |
```

PROGRAM 3

AIM:

Write a program to find LCM and GCD of two numbers

PROGRAM:

```
a=int(input("Enter a number"))
b=int(input("Enter another number"))

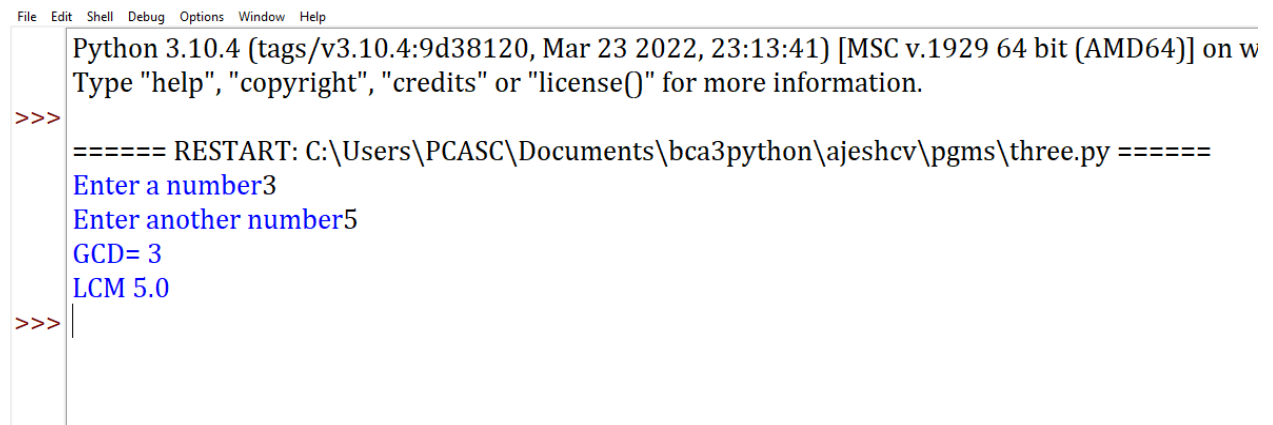
for i in range(1,min(a,b)+1):
    if a%i==0 & b%i==0:
        g=i

print("GCD=",g)

lcm=(a*b)/g

print("LCM",lcm)
```

OUTPUT:

A screenshot of a Python IDE window. The title bar shows 'File Edit Shell Debug Options Window Help'. The main text area displays the output of a Python script. It starts with the Python version and architecture: 'Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (AMD64)] on w'. Below this, it says 'Type "help", "copyright", "credits" or "license()" for more information.' The prompt '>>>' is followed by a restart message: '==== RESTART: C:\Users\PCASC\Documents\bca3python\ajeshcv\pgms\three.py ====='. Then, the user inputs '3' for 'Enter a number' and '5' for 'Enter another number'. The program outputs 'GCD= 3' and 'LCM 5.0'. The prompt '>>>' is shown again at the bottom.

```
File Edit Shell Debug Options Window Help
Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (AMD64)] on w
Type "help", "copyright", "credits" or "license()" for more information.
>>>
==== RESTART: C:\Users\PCASC\Documents\bca3python\ajeshcv\pgms\three.py =====
Enter a number3
Enter another number5
GCD= 3
LCM 5.0
>>>
```

PROGRAM 4

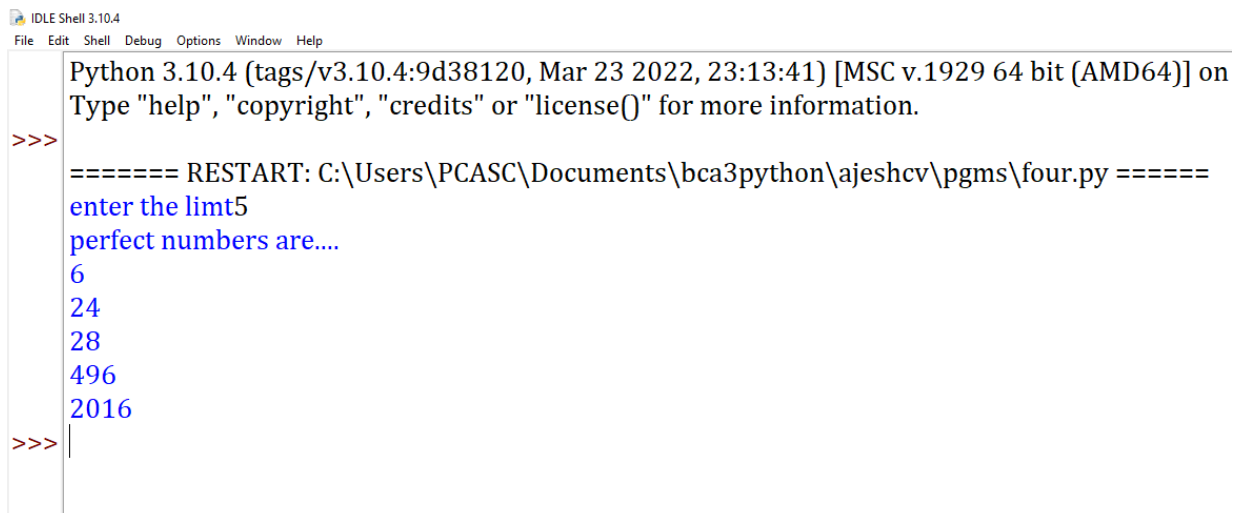
AIM:

Write a program to generate first n perfect numbers

PROGRAM:

```
def check_perfect(n):  
    count,i=0,0  
  
    print("perfect numbers are....")  
  
    while count<n:  
        i=i+1  
  
        sum=0  
  
        for item in range(1,i):  
            if i%item==0:  
                sum=sum+item  
  
            if i==sum:  
                print(i)  
                count=count+1  
  
n=int(input("enter the limit"))  
  
check_perfect(n)
```

OUTPUT:



```
IDLE Shell 3.10.4  
File Edit Shell Debug Options Window Help  
Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (AMD64)] on  
Type "help", "copyright", "credits" or "license()" for more information.  
>>>  
===== RESTART: C:\Users\PCASC\Documents\bca3python\ajeshcv\pgms\four.py =====  
enter the limit5  
perfect numbers are....  
6  
28  
496  
2016  
>>>
```

PROGRAM 5

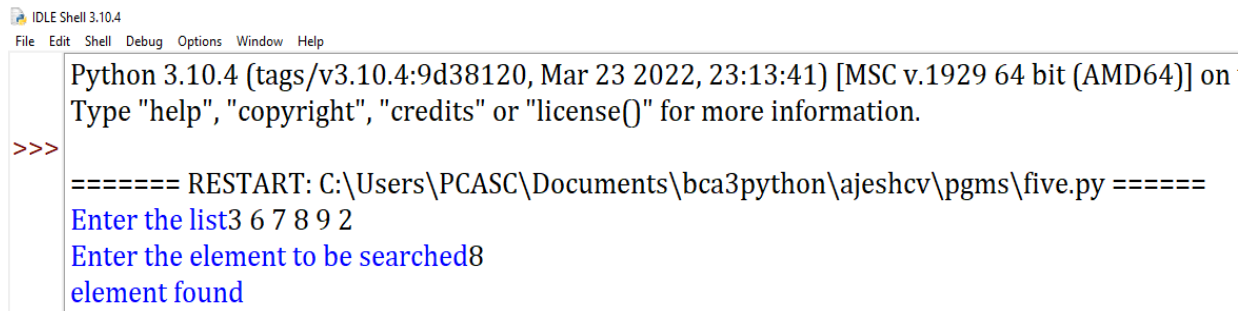
AIM:

Write a program to perform a binary search

PROGRAM:

```
a=input("Enter the list")
li=list(map(int,a.split()))
li.sort()
e=int(input("Enter the element to be searched"))
high=len(li)-1
low=0
while low<=high:
    mid=(low+high)//2
    if e==li[mid]:
        print("element found")
        break
    elif e>li[mid]:
        low=mid+1
    else:
        high=mid-1
if low>high:
    print("element not found")
```


OUTPUT:



```
IDLE Shell 3.10.4
File Edit Shell Debug Options Window Help

Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (AMD64)] on
Type "help", "copyright", "credits" or "license()" for more information.

>>>

===== RESTART: C:\Users\PCASC\Documents\bca3python\ajeshcv\pgms\five.py =====
Enter the list3 6 7 8 9 2
Enter the element to be searched8
element found
```

PROGRAM 6

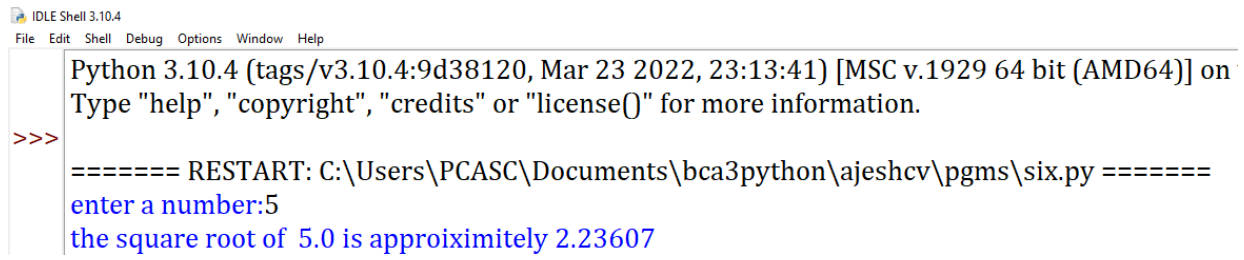
AIM:

Write a program to find the square root of a number using bisection search

PROGRAM:

```
def find_square_root(number,precision=0.00001):  
    if number<0:  
        raise ValueError("square root is not defined for negative number")  
  
    if number==0:  
        return 0  
  
    low=0  
    high=max(1,number)  
    guess=(low+high)/2  
    while abs(guess**2-number)>precision:  
        if guess**2<number:  
            low=guess  
        else:  
            high=guess  
        guess=(low+high)/2  
  
    return guess  
  
number=float(input("enter a number:"))  
precision=0.00001  
result=find_square_root(number,precision)  
print(f"the square root of {number} is approximately {result:.5f}")
```

OUTPUT:



The screenshot shows the IDLE Shell 3.10.4 interface. The title bar reads 'IDLE Shell 3.10.4'. The menu bar includes 'File', 'Edit', 'Shell', 'Debug', 'Options', 'Window', and 'Help'. The main text area displays the following output:

```
Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (AMD64)] on  
Type "help", "copyright", "credits" or "license()" for more information.  
>>>  
===== RESTART: C:\Users\PCASC\Documents\bca3python\ajeshcv\pgms\six.py =====  
enter a number:5  
the square root of 5.0 is approximately 2.23607
```

PROGRAM 7

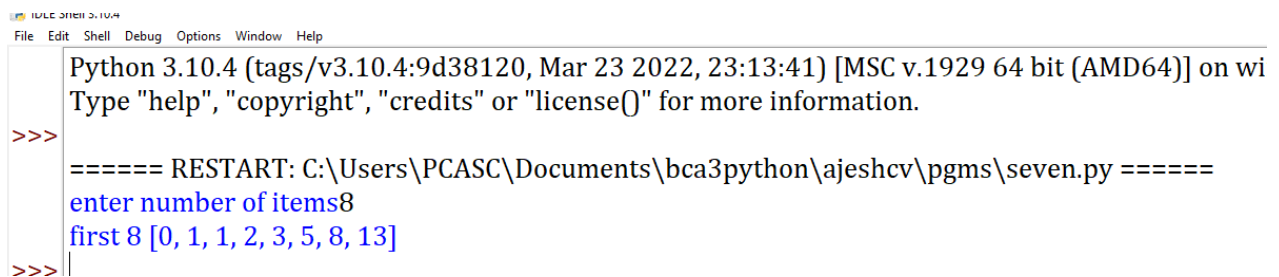
AIM:

Write a program to generate Fibonacci series using recursion

PROGRAM:

```
def fibonacci(n):  
    if n<=0:  
        return  
    elif n==1:  
        return 0  
    elif n==2:  
        return [0,1]  
    else:  
        fib_list=fibonacci(n-1)  
        fib_list.append(fib_list[-1]+fib_list[-2])  
        return fib_list  
  
n=int(input("enter number of items"))  
  
fib_series=fibonacci(n)  
  
print("first",n,fib_series)
```

OUTPUT:



The screenshot shows a Python IDE window titled 'IDLE 3.10.4'. The menu bar includes File, Edit, Shell, Debug, Options, Window, and Help. The main text area displays the following output:

```
Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (AMD64)] on wi  
Type "help", "copyright", "credits" or "license()" for more information.  
>>>  
===== RESTART: C:\Users\PCASC\Documents\bca3python\ajeshcv\pgms\seven.py =====  
enter number of items8  
first 8 [0, 1, 1, 2, 3, 5, 8, 13]  
>>>|
```

PROGRAM 8

AIM:

Write a program to perform merge sort

PROGRAM:

```
def merge(left, right):  
    output = []  
    i = j = 0  
    while i < len(left) and j < len(right):  
        if left[i] < right[j]:  
            output.append(left[i])  
            i += 1  
        else:  
            output.append(right[j])  
            j += 1  
    output.extend(left[i:])  
    output.extend(right[j:])  
    return output  
  
def mergesort(lis):  
    if len(lis) <= 1:  
        return lis  
    mid_point = len(lis) // 2  
    left_partition = mergesort(lis[:mid_point])  
    right_partition = mergesort(lis[mid_point:])  
    return merge(left_partition, right_partition)
```

```
li = []

n = int(input("Enter the number of elements in the list: "))

for i in range(n):

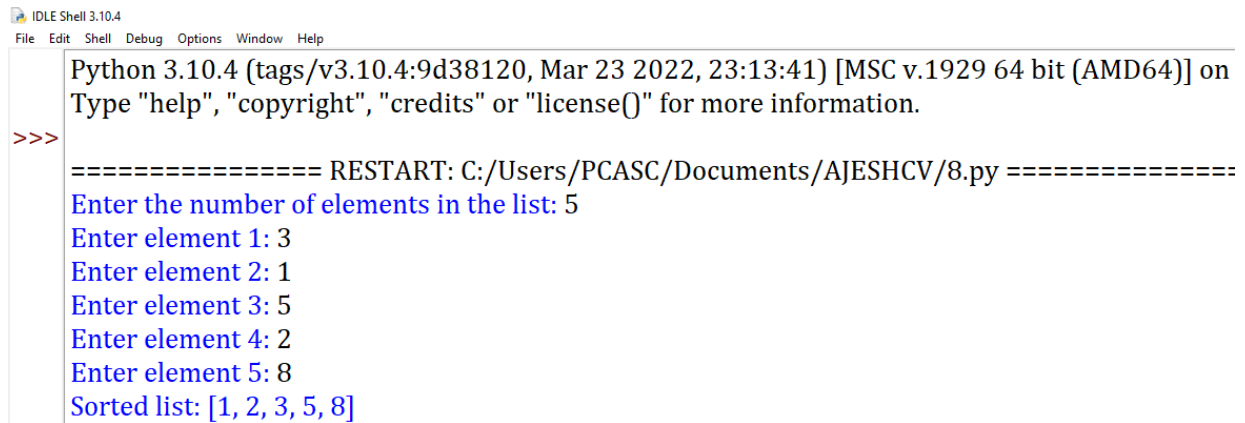
    a = int(input(f"Enter element {i+1}: "))

    li.append(a)

sorted_list = mergesort(li)

print("Sorted list:", sorted_list)
```

OUTPUT:



```
IDLE Shell 3.10.4
File Edit Shell Debug Options Window Help
Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (AMD64)] on
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/PCASC/Documents/AJESHCV/8.py =====
Enter the number of elements in the list: 5
Enter element 1: 3
Enter element 2: 1
Enter element 3: 5
Enter element 4: 2
Enter element 5: 8
Sorted list: [1, 2, 3, 5, 8]
```

PROGRAM 9

AIM:

Write a program which reads the context of a file and copy context to another File after changing the entire letter to uppercase. Exception should be handled

PROGRAM:

try:

```
f1=open("luffy.txt.txt","r")
f2=open("zoro.txt.txt","w")
lines=f1.read()
print("Before copying",lines)
lines=lines.upper()
f2.write(lines)
print("After cpoying",lines)
f1.close()
f2.close()
```

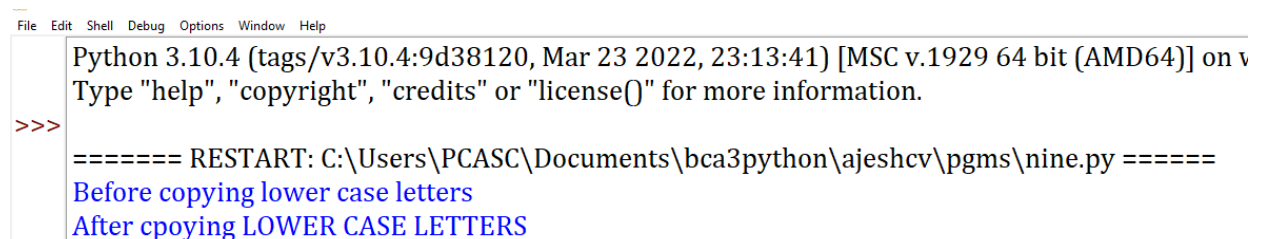
except FileNotFoundError:

```
print("file not found")
```

except IOError:

```
print("IO error")
```

OUTPUT:



```
File Edit Shell Debug Options Window Help
Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (AMD64)] on v
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:\Users\PCASC\Documents\bca3python\ajeshcv\pgms\nine.py =====
Before copying lower case letters
After cpoying LOWER CASE LETTERS
```

PROGRAM 10

AIM:

Create a simple log in window using Tkinter

PROGRAM:

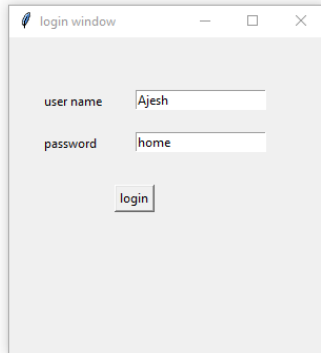
```
def btn_clk():  
    if pwd.get()=='home':  
        newwin=Tk()  
        newwin.geometry('400x300')  
        newwin.title('Homepage window')  
        greet="Hello,"+nm.get()+",login succesfull"  
        Label(newwin,text=greet,fg='Blue').place(x=100,y=100)  
    else:  
        Label(win,text='Inavlid username and/or password',fg='Blue').place(x=40,y=200)  
  
from tkinter import Tk, Label, Button, Entry, StringVar  
  
win=Tk()  
win.geometry("300x300")  
win.title('login window')  
  
nm=StringVar()  
pwd=StringVar()  
  
name=Label(win,text="user name").place(x=30,y=50)  
password=Label(win,text="password").place(x=30,y=90)  
  
submitbtn=Button(win,text="login",activebackground="yellow",activeforeground="red",  
command=btn_clk).place(x=100,y=140)  
  
e1=Entry(win,textvariable=nm).place(x=120,y=50)  
e2=Entry(win,textvariable=pwd).place(x=120,y=90)  
  
win.mainloop()
```


OUTPUT

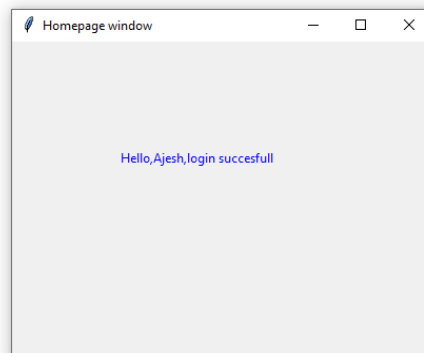
```
*IDLE Shell 3.10.4*
File Edit Shell Debug Options Window Help
Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (AMD64)] on
Type "help", "copyright", "credits" or "license()" for more information.
```

```
>>>
```

```
===== RESTART: C:\Users\PCASC\Documents\bca3python\ajeshcv\pgms\ten02.py =====
```



A screenshot of a 'login window' with a light gray background. It contains two text input fields: 'user name' with the text 'Ajesh' and 'password' with the text 'home'. Below the password field is a button labeled 'login'.



PROGRAM 11

AIM:

Write a program to perform the following.

- a) Create table student with fields name, sex, roll no, mark
- b) Insert some rows into table
- c) Delete a student with a given roll no
- d) Update the marks of all students by adding 2 marks
- e) Display the details of a student with a given roll no

PROGRAM a:

```
import pymysql as ps
```

```
try:
```

```
    db = ps.connect(host="localhost", user="root", password="",  
database="databaseforbcanew")
```

```
except Exception as e:
```

```
    print("Database connection error:", e)
```

```
    exit()
```

```
try:
```

```
    cursor = db.cursor()
```

```
    sql = "CREATE TABLE students(name VARCHAR(20), sex VARCHAR(20), regno INT,  
mark INT)"
```

```
    cursor.execute(sql)
```

```
    db.commit()
```

```
    print("Table created successfully.")
```

```
except Exception as e:
```

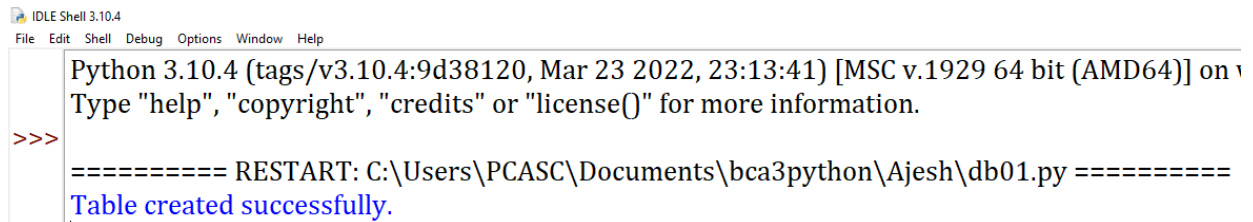
```
    print("Table creation error:", e)
```

```
    db.rollback()
```

```
finally:
```

```
    db.close()
```

OUTPUT:



```
IDLE Shell 3.10.4
File Edit Shell Debug Options Window Help
Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (AMD64)] on v
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:\Users\PCASC\Documents\bca3python\Ajesh\db01.py =====
Table created successfully.
```

PROGRAM b:

```
import pymysql as ps
```

```
try:
```

```
    db = ps.connect(host="localhost", user="root", password="",
database="databaseforbcanew")
```

```
except Exception as e:
```

```
    print("Database connection error:", e)
```

```
    exit()
```

```
try:
```

```
    cursor=db.cursor()
```

```
    while(True):
```

```
        name=input("enter the name of the student:")
```

```
        gender=input("enter the gender of the student:")
```

```
        regno=int(input("enter the resgister number of the student:"))
```

```
        mark=int(input("enter the mark of the student:"))
```

```
        ins_sql = "INSERT INTO students VALUES('{}', '{}', {}, {})".format(name, gender,
regno, mark)
```

```
        cursor.execute(ins_sql)
```

```
        db.commit()
```

```
        print("one view inserted")
```

```
        resp=input("Do you want to continue one more record(y/n):")
```

```
        resp=resp.lower()
```

```
if(resp!='y'):
```

```
    break
```

```
except:
```

```
    print("can not insert a row")
```

OUTPUT:

```
mysql> select * from students;
+-----+-----+-----+-----+
| name   | sex   | regno | mark |
+-----+-----+-----+-----+
| Ajesh  | Male  | 1     | 60   |
| Akshith | Male  | 2     | 46   |
| Mridul | Male  | 3     | 45   |
| Adith  | Male  | 4     | 99   |
| Aswima | Female | 5     | 55   |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

PROGRAM C:

```
import pymysql as ps
```

```
try:
```

```
    db = ps.connect(host="localhost", user="root", password="",
database="databaseforbca")
```

```
except Exception as e:
```

```
    print("Database connection error:", e)
```

```
    exit()
```

```
try:
```

```
    cursor = db.cursor()
```

```
    regno = int(input("Enter the roll number of the student to delete: "))
```

```
    delete_query = "DELETE FROM students WHERE regno = %s"
```

```
    cursor.execute(delete_query, (regno,))
```

```

db.commit()

if cursor.rowcount > 0:

    print(f"Student with roll number {regno} deleted successfully.")

else:

    print(f"No student found with roll number {regno}.")

cursor.close()

db.close()

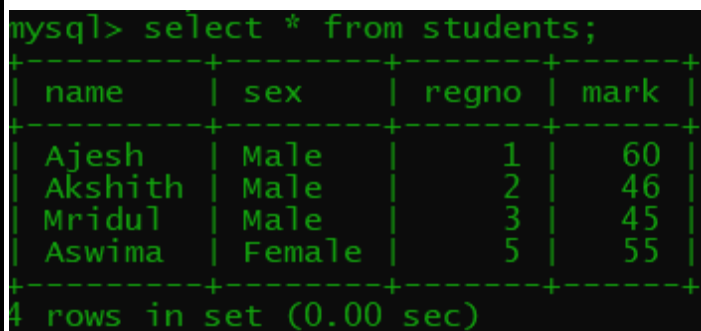
print("Connection closed.")

except Exception as e:

    print(f"Error: {e}")

```

OUTPUT:



```

mysql> select * from students;
+-----+-----+-----+-----+
| name   | sex   | regno | mark |
+-----+-----+-----+-----+
| Ajesh  | Male  | 1     | 60   |
| Akshith | Male  | 2     | 46   |
| Mridul  | Male  | 3     | 45   |
| Aswima  | Female | 5     | 55   |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

```

PROGRAM D:

```

import pymysql as ps

try:

    db = ps.connect(host="localhost", user="root", password="",
database="databaseforbcanew")

except Exception as e:

    print("Database connection error:", e)

```

```
exit()
```

```
try:
```

```
    cursor = db.cursor()
```

```
    update_query = "UPDATE students SET mark = mark + 2"
```

```
    cursor.execute(update_query)
```

```
    db.commit()
```

```
    updated_count = cursor.rowcount
```

```
    print(f"{updated_count} student mark updated successfully.")
```

```
    cursor.close()
```

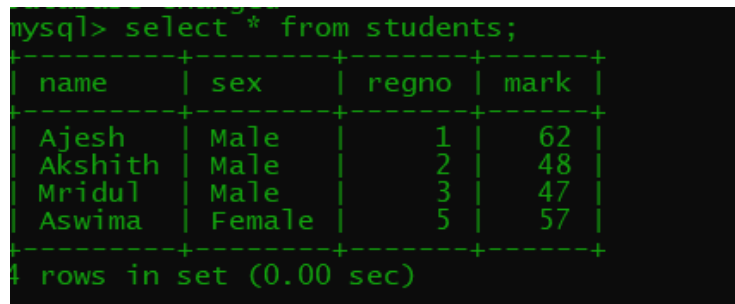
```
    db.close()
```

```
    print("Connection closed.")
```

```
except Exception as e:
```

```
    print(f"Error: {e}")
```

OUTPUT:



```
mysql> select * from students;
```

name	sex	regno	mark
Ajesh	Male	1	62
Akshith	Male	2	48
Mridul	Male	3	47
Aswima	Female	5	57

```
4 rows in set (0.00 sec)
```

PROGRAM E:

```
import pymysql as ps
```

```
try:
```

```
    db = ps.connect(host="localhost", user="root", password="",  
    database="databaseforbcanew")
```

```
except Exception as e:
```

```
print("Database connection error:", e)

exit()

try:

    cursor = db.cursor()

    regno = int(input("Enter the roll number of the student to display:"))

    select_query = "SELECT * FROM students WHERE regno = %s"

    cursor.execute(select_query, (regno,))

    student = cursor.fetchone()

    if student:

        print("Student details:")

        print(f"Name: {student[0]}")

        print(f"Sex: {student[1]}")

        print(f"RegNo: {student[2]}")

        print(f"Marks: {student[3]}")

    else:

        print(f"No student found with roll number {regno}.")

except ps.Error as error:

    print(f"Error: {error}")

finally:

    if db:

        db.close()

    print("MySQL connection closed.")
```

OUTPUT:

```
Python 3.10.4 (tags/v3.10.4:9d38120, Mar 23 2022, 23:13:41) [MSC v.1929 64 bit (
Type "help", "copyright", "credits" or "license()" for more information.

>>>

===== RESTART: C:\Users\PCASC\Documents\bca3python\Ajesh\db05.py =
Enter the roll number of the student to display:1
Student details:
Name: Ajesh
Sex: Male
RegNo: 1
Marks: 62
MySQL connection closed.
```


PROGRAM 12

AIM:

Create a plot for the mathematical function x^2 the title of plot and the axes should be labeled

PROGRAM:

```
import matplotlib.pyplot as plt

import numpy as np

x=np.arange(-50,50)

y=x*x

plt.plot(x,y)

plt.xlabel('x values from -50 to 50')

plt.ylabel('x^2')

plt.title("y='x^2'function")

plt.show()
```

OUTPUT:

